Exam 1 review

09/30/2026

Single phase power

- RMS phasors

$v(t) = V_m \cos(\omega t + \theta_v) \rightarrow \bar{V} = \frac{V_m}{\sqrt{2}} \angle \theta_v$ (V_{RMS})

if neg., fix it by $\pm 180°$ shift

if sin, $\sin x = \cos(x - 90°)$

- $P, Q, \bar{S}$ -?

$P = Re\{\bar{V}\bar{I}^*\}$ [W]

$Q = Im\{\bar{V}\bar{I}^*\}$ [VAR]

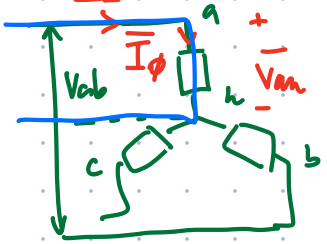
$\bar{S} = P + jQ = \bar{V} \cdot \bar{I}^*$ [VA]

- Power triangle:

Lagging PF (inductive load)	Leading PF (capacitive load)	Unity PF (resistive load)
Current lags voltage	Current leads voltage	Current and voltage are in phase
$\bar{V}$, $\bar{I}$ $\Rightarrow \theta = \theta_v - \theta_i > 0$	$\bar{I}$, $\bar{V}$ $\Rightarrow \theta = \theta_v - \theta_i < 0$	$\bar{I}$, $\bar{V}$ $\Rightarrow \theta = \theta_v - \theta_i = 0$
S, θ, P; $Q > 0$	P, θ, S; $Q < 0$	$P = S$; $Q = 0$

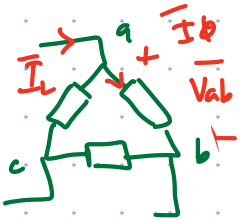
Three-phase power

Y-Y



- $V_L = V_{ab} = \sqrt{3} V_{an} = \sqrt{3} V_{\phi,Y}$
- $I_L = I_{\phi,Y}$

Y-Δ



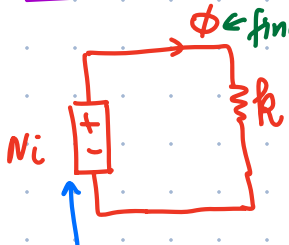
- $V_L = V_{ab} = V_{\phi,\Delta}$
- $I_L = \sqrt{3} I_{\phi,\Delta}$

$\bar{S}_{3\phi} = 3\bar{S}_\phi = 3\bar{V}_\phi \cdot \bar{I}_\phi^*$

$\Delta \rightarrow Y$: $\bar{Z}_Y = \frac{\bar{Z}_\Delta}{3}$

$PF_\phi = PF_{3\phi}$ (doesn't change)

Magnetic equivalent circuits



polarity: ϕ flows out of "+" into "−"

$\lambda = Li = N\phi$

$v = d\lambda/dt = L\frac{di}{dt}$

$\mathcal{R} = \frac{l}{\mu_r \mu_0 A}$ $(\mu \rightarrow \infty, \mathcal{R} \rightarrow 0)$

if any of these changes, we need to introduce new reluctance

Fringing: $A_g \uparrow$

Coupled coils

Dot convention: current entering dots throw flux in the same direction.

- Put dots properly
- Check if actual case matches reference model
  - yes → apply ☺
  - no → fix i and/or v → apply ref. model → substitute back

① Spring 2024, M1, pr 2

Problem 2 (25 Points)

A 3-phase, 208 V (line-line), Wye-connected source is supplying two 3-phase loads connected in parallel:

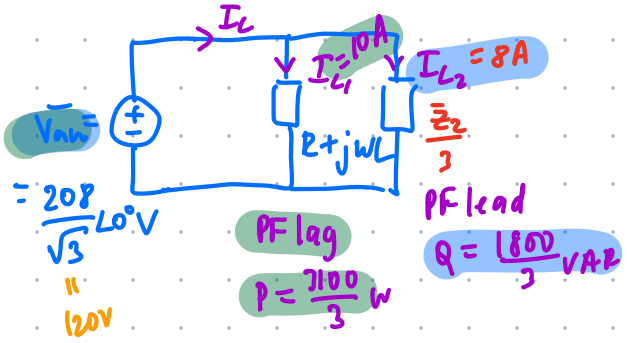
1. Wye-connected, inductive, draws 10 A phase current and 3100 W of real power (3-phase)
2. Delta-connected, it has per-phase impedance $\bar{Z}_2$, draws 8 A line current and 1800 VARs of reactive power (3-phase) at a leading power factor.

PF lag, = line, capacitive

a) Find the total real power (3-phase) supplied by the source.

P_{tot} = $P_1 + P_2$ = 5350 W (3100W) & (2250W) Unit=____

Per-phase eq. circuit:



Load 1:

$P = V_{an} \cdot I_{L_1} \cdot PF \Rightarrow PF = \frac{3100/3}{120 \cdot 10} = 0.86$

$\theta = \cos^{-1}(0.86) = 30.6°$

$\theta = \theta_v - \theta_i \Rightarrow \theta_i = -30.6°$

load 2: $S^2 = P^2 + Q^2 \Rightarrow P = \sqrt{S^2 - Q^2} = 750$ W

↑ ($V_{an} \cdot I_{L_2}$)

Load 2: $PF = \frac{750}{120 \cdot 8} = 0.78$ lead

$\theta_i = \cos^{-1}(0.78) = 38.7°$

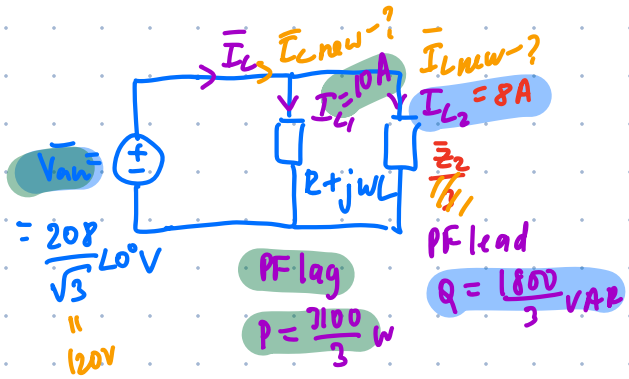
$\bar{I}_{L_2} = 8 \angle 38.7°$ A

b) Compute the total line current (magnitude) being supplied by the source.

I_L = $|\bar{I}_{L_1} + \bar{I}_{L_2}| = 14.84$ A Unit=____

c) Find per-phase impedance of the second load, $\bar{Z}_2$

$\bar{Z}_2$=_____Unit=____



$\Rightarrow \bar{I}_{L_2} = \frac{\bar{V}_{an}}{(\frac{\bar{Z}_2}{3})} \Rightarrow \bar{Z}_2 = \frac{3 \cdot 120\angle 0°}{8\angle 38.7°} = 45\angle -38.7°\ \Omega$

$\bar{I}_{L_{2new}} = \frac{120\angle 0°}{45\angle -38.7°} = 2.66\angle 38.7°\ A$

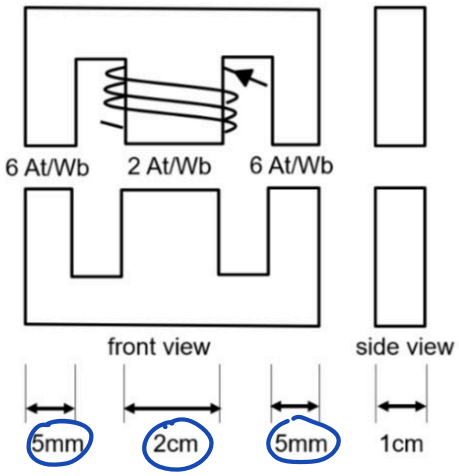
d) Now we reconfigure the second load to be connected in Wye (instead of Delta, as originally indicated), while keeping its per-phase impedance $\bar{Z}_2$ the same as computed in part c. What is the new total line current (magnitude) being supplied by the source?

$I_{L,new}$ = 11.2 A Unit=____

② Fall 2023, M1, pr 3

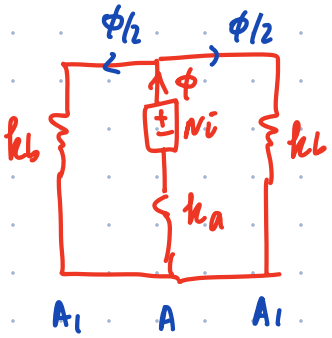
Problem 3 (25 Points)
Note: Parts **a)**, **b)** and **c)** all refer to the same structure described below.

Consider the iron structure shown. Assume that $\mu = \infty$ in the iron. The coil has $N = 12$ turns.



a) Draw the magnetic circuit for the structure. Please take care to orient the polarities correctly. Label the mmfs. Use the magnetic circuit to compute the inductance of the structure.

a)



$\lambda = L \cdot i = N\Phi \Rightarrow L = \frac{N^2}{\mathcal{R}_{tot}} = \frac{144}{5}\,H$

$\Phi = \frac{Ni}{\mathcal{R}_{tot}}$ $\mathcal{R}_{tot} = \mathcal{R}_a + \frac{\mathcal{R}_b \cdot \mathcal{R}_b}{\mathcal{R}_b + \mathcal{R}_b} = \mathcal{R}_a + \frac{\mathcal{R}_b}{2} = 5\,\frac{A \cdot t}{Wb}$

b) The flux density in each of three legs must be kept at equal or below $B = 0.1$ T, or else the core will saturate. What is the <u>maximum current</u> i_{max} that we can drive the coil without causing saturation?

$B \leq 0.1\,T$ each leg

$\phi = B \cdot A \Rightarrow \frac{\phi}{A} \leq 0.1 \Rightarrow \frac{N \cdot i_{max}}{\mathcal{R}_{tot} \cdot A} \leq 0.1$

$\frac{\phi}{2 \cdot A_1} \leq 0.1 \Rightarrow \frac{N i_{max}}{2\mathcal{R}_{tot} \cdot A_1} \leq 0.1$

↑ bottleneck

$i_{max} = 4.16\,\mu A$

c) The reluctances stated in the diagram were computed neglecting fringing. If the effects of fringing were instead accounted for, how would your answer to part b) change? Would the maximum current i_{max} without causing saturation increase or decrease? Explain.

↓ $A_g \uparrow \Rightarrow \mathcal{R}_g \downarrow \Rightarrow \mathcal{R}_{tot} \downarrow \Rightarrow i_{max} = \frac{2\mathcal{R}_{tot} \cdot A_1 \cdot 0.1}{N^2}$ ↓

Note: Credit will <u>only</u> be awarded for the explanation. No credit will be awarded for the correct choice.

Increase | Decrease | Stay same | Insufficient info

Explanation: